# RETATRUTIDE (LY3437943)

Investigational GIP / GLP-1 / Glucagon Triple Receptor Agonist
Scientific Literature Review for Qualified Laboratory Researchers

## Compound Identity and Development Status

Retatrutide (chemical development code **LY3437943**) is a synthetic single-chain peptide analog designed as a simultaneous agonist of three G-protein-coupled receptors: the glucose-dependent insulinotropic polypeptide (GIP) receptor, the glucagon-like peptide-1 (GLP-1) receptor, and the glucagon receptor. The molecule was discovered and is being developed by Eli Lilly and Company. As of August 2026 it remains an **investigational new drug**. No New Drug Application or Biologics License Application has been approved by the U.S. Food and Drug Administration or any other national regulator. Lilly has publicly indicated an intent to submit a BLA in the first quarter of 2027. Until approval is granted, the compound is legally available only under authorized clinical-trial protocols.

## Published Receptor Pharmacology (Literature Summary)

| GLP-1 Receptor | GIP Receptor | Glucagon Receptor |
|---|---|---|
| Published studies associate GLP-1R agonism with delayed gastric emptying, incretin-mediated insulin secretion, and reduced caloric intake in experimental models. | GIPR agonism has been reported to enhance post-prandial insulin response and to influence adipose-tissue lipid handling in preclinical and clinical literature. | Glucagon-receptor agonism has been associated in published work with increased energy expenditure and hepatic lipid mobilization in research settings. |

## Selected Peer-Reviewed and Sponsor-Reported Clinical Data

The following figures are taken from publicly disclosed trial publications and sponsor topline releases. They describe outcomes observed under controlled clinical-trial conditions with pharmaceutical-grade material and do not constitute claims about any research-chemical product.

**Phase 2 obesity trial** — Jastreboff et al., *N Engl J Med* 2023;389:514-526. 338 adults. At 48 weeks, least-squares mean body-weight change: –8.7% (1 mg), –17.1% (4 mg combined), –22.8% (8 mg combined), –24.2% (12 mg) versus –2.1% placebo.

*Jastreboff AM et al. Triple-Hormone-Receptor Agonist Retatrutide for Obesity — A Phase 2 Trial. NEJM 2023. DOI: 10.1056/NEJMoa2301972*

**Phase 3 TRIUMPH-1** (obesity, no T2D; n=2,339; 80 weeks; sponsor topline May 2026 / ADA 2026). Mean body-weight change vs placebo (–2.2%): 4 mg –19.0%; 9 mg –25.9%; 12 mg –28.3%. At 12 mg, 45.3% of participants reached ≥30% weight reduction. Pre-specified 104-week extension subset: –30.3% at 12 mg.

*Eli Lilly TRIUMPH-1 topline and ADA 2026 symposium materials. ClinicalTrials.gov NCT05929066.*

**Phase 3 TRIUMPH-2** (obesity or overweight + T2D; 80 weeks). Mean weight change at 12 mg –20.8% (≈49.6 lb) with A1C reduction up to 1.6 percentage points. **TRIUMPH-3** (severe obesity + established CVD): up to –22.6% (≈55.8 lb) at 12 mg.

*Eli Lilly press release, 23 July 2026. Additional Phase 3 programs (sleep apnea, osteoarthritis nested in TRIUMPH-1) also reported positive secondary endpoints.*

## Regulatory and Legal Status (United States, August 2026)

- Retatrutide is **not** an FDA-approved drug and is **not** a component of any approved drug product.
- FDA has stated that retatrutide **cannot lawfully be used in compounding** under FD&C Act sections 503A or 503B.
- Introduction into interstate commerce of unapproved new drugs is prohibited (21 U.S.C. §§ 331(d), 355(a)).
- FDA warning letters (2024–2026) have treated website language describing appetite, weight, insulin, or metabolic effects as evidence of intended drug use, even when products are labeled “research use only.”
- In August 2026 Eli Lilly filed federal lawsuits against multiple U.S. sellers (including research-chemical vendors) alleging illegal marketing of products purporting to contain retatrutide.